

Foundation of a relativistic rigid body rotation theory

Stefan Catheline

LabTAU, INSERM, Centre Léon Bérard, Université Lyon 1, Univ Lyon, F-69003, LYON, France

stefan.catheline@inserm.fr

ABSTRACT

Does the mathematical straight line actually exist in our physical world? Since the observation of circular trajectories at all scales is undeniable, this manuscript chooses to answer “no” to that question and therefore presents a theory of rotation. From the concept of relativistic rigid body rotation introduced in a recent publication, we present in this manuscript the theoretical foundation of a mechanics entirely built on the base of a dynamic space trihedron composed of rotation, velocity and distance vectors.

Introduction

The starting point of the rotation theory presented in this manuscript is the relativist description of a physical phenomenon that, like a sea serpent, repeatedly returned several times to question physicists: rigid body rotation. It was indeed omnipresent right at the beginning of modern sciences through, for example, vortex theories of René Descartes or William Kelvin. More recently, it was proposed to Albert Einstein by Kurt Gödel [1] as a universe model. It was also the main support used by Lev Landau [2] to introduce general relativity. However, rigid body rotation presents a big flaw of connection to infinity: at large enough distances, the speed can reach and overcome light speed. This difficulty can be avoided, and this is the main idea of the following developments, the basement on which the whole theoretical construction is built: rigid body rotation has a horizon at finite distance [3]. This latter property of spatial confinement [4] is the central constitutive element of relativistic rigid body rotation that shall now be described.

Background of relativistic rigid body rotation

The relativistic rigid body rotation has already been described in a previous publication [3]. Some useful properties are quickly recalled here. First, as any rigid body rotation, it is composed of an angular rotation $\vec{\omega}$, a velocity $\vec{v}$ and a distance radius $\vec{r}_0$ from the center O. These three vectors are connected by the equation $\vec{v} = \vec{\omega} \wedge \vec{r}_0$, figure 1. The entire originality lies in the fact that the physical path followed by a light ray emitted from O is not the straight line $\vec{r}_0$. Instead, this radial path imposed by the Coriolis acceleration, follows a simple circle called “space circle”. The arc of space circle to the satellite position is deduced from the angle rotation Ω and the half radius vector of the rotation $\frac{\vec{R}}{2}$:

$$\vec{r} = \vec{\Omega} \wedge \frac{\vec{R}}{2} \quad (1).$$

Through transverse Doppler effect, the space circle has been described in 1963 [5]. It was then confirmed with metric tensor approach borrowed to relativity [6]. The angular distortion θ between the direction of the velocity and of the space circle tangent results in radial v_r and

orthoradial v_ϕ component of velocity. It thus follows that acceleration as well presents new components. The complete expression of Eulerian acceleration can be expressed as follows:

$$\vec{a} = \frac{\partial \vec{v}}{\partial t} + \vec{\omega} \wedge \vec{v}_\phi + 2\vec{\omega} \wedge \vec{v}_r \quad (2).$$

At last, the relativistic rotation fully satisfies the space-time Lorentz transformation. Relativistic parameters follow the geometry of figure 1 through the equations $\cos\theta = \frac{r_0}{R} = \frac{v}{c}$ on one side, and $\sin\theta = \sqrt{1 - \frac{v^2}{c^2}}$ on the other side.

When the satellite velocity v is small compared to that of light c , the classic case of rigid body rotation is recovered. When on the contrary, the velocity is close to the speed of light, the satellite reaches its maximum radius R on the rigid body rotation horizon. Again, the space circle tangent experiences an angular distortion θ with respect to the velocity direction which prevents any speed value bigger than the light speed limit c . By construction, velocity values are bounded: the first of all universal constant of physics c , is thus a consequence of the finite geometrical nature of the relativistic rigid body rotation. This novel approach conveys a simple and straightforward interpretation of the spin as the expression of horizon [4].

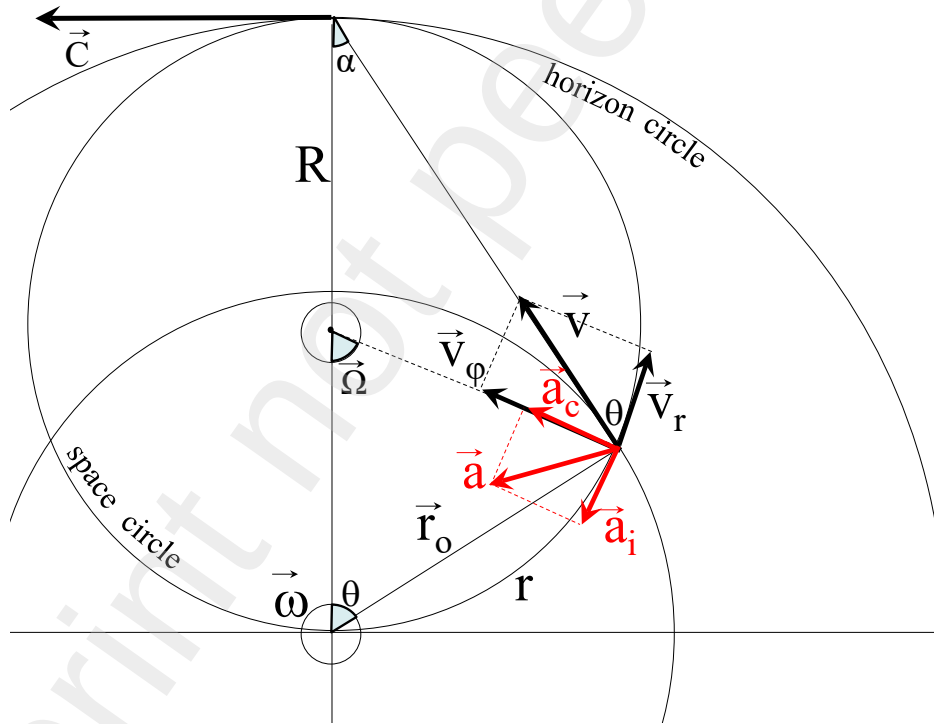


Fig. 1 : Relativistic rigid body rotation. Three circles are necessary for its construction: the speed circle of radius r_0 , the space circle of radius $R/2$ and the rotation circle of radius R defining the horizon of the rotation.

Cinematic postulate and fundamental constants

The departure postulate introduced up to now stipulates that any trajectory is necessarily under the influence of at least one relativistic rotation.

Kinematic postulate

$\omega = \frac{v}{r_0} = \frac{c}{R} > 0$ or $\vec{v} = \vec{\omega} \wedge \vec{r}_0 > \vec{0}$, **any trajectory results from a relativistic rotation (3).**

This ‘all-rotation’ postulate holds instead of the first law of inertia (Galileo-Newton). It is a purely kinematic postulate and also geometric as it defines the curved nature of space itself, Figure 1. The static Galilean space (x,y,z) is replaced by a dynamic space $(\vec{\omega}, \vec{r}, \vec{v})$. Just like the quadridimensional Minkowsky space (x,y,z,t), the dynamic space $(\vec{\omega}, \vec{r}, \vec{v})$ includes time within its base but in an implicit manner. In opposition to its classical *alter ego* the Galilean referential, the relativistic rotation referential involves a finite region of space only. Therefore, the rotation presents by construction a maximum velocity: this first universal constant c is thus a direct consequence of the kinematic postulate, and of the geometric nature of relativistic rigid body rotation. By pushing the logic of symmetry between v, r and ω further, it becomes necessary to accept that, if the velocity v has a maximum value c, then distance and rotation also admit maxima $r_{\max}$ and $\omega_{\max}$. Then, three additional constants immediately follow equation (3): from c and $\omega_{\max}$ the minimum distance $r_{\min}$ is deduced, from c and $r_{\max}$ the minimum rotation $\omega_{\min}$ emerges and finally from $\omega_{\min}$ and $r_{\min}$, the minimum velocity $v_{\min}$ is defined. Three independent universal constants are needed to build a complete set of six quantities in which null or infinite values are prohibited. Another general argument can justify the prohibition of null values: a 3D-space is described by unit vectors $\vec{u}_x$, $\vec{u}_y$ and $\vec{u}_z$ that cannot be zero by definition except in special plan-space (2D), line-space (1D) or point-space (0D). The same logic applies for relativistic rigid body rotation $(\vec{\omega}, \vec{r}, \vec{v})$. Null vector would annihilate any notion of trajectory and space as well as would cancel the kinematic postulate. The cinematic values associated to the six boundary constants are partly inspired by actual scientific knowledge; it is the speed of light, the radius of universe, the Planck rotation, the Hubble constant and the Planck distance. To summarize, these values with their international units are gathered in a table:

$v_{\max} = c$	$r_{\max} = R_U$	$\omega_{\max} = \omega_{pl}$	$v_{\min}$	$r_{\min} = r_{pl}$	$\omega_{\min} = H_0$
$3 \cdot 10^8 \text{ m.s}^{-1}$	10^{26} m	$6 \cdot 10^{43} \text{ rad.s}^{-1}$	$2 \cdot 10^{-53} \text{ m.s}^{-1}$	$1.6 \cdot 10^{-35} \text{ m}$	$210^{-18} \text{ rad.s}^{-1}$

Table 1 : Universal constants and boundary values of physics

Ether hypothesis, first physical law and elementary particle

The ether hypothesis introduces physics within the kinematics of relativistic rotation by describing the nature of what is in motion: a uniform distribution of mass density, a kind of ether for which the origin and properties are about to be described. This inherent mass density of space confers to the rigid body rotation ω a moment of inertia $J = mr_0^2$, a moment of mass $\Gamma = mr_0$, and simply speaking, a mass m. This mass distribution is now simplified into a simple point mass m_0 in rotation. Let us calculate the total energy rotation, that is to say the work needed for a given force to bring a motionless particle having a rest mass m_0 from the rotation center O to its horizon R. A classical approach based on inertial force $F = m_0 \frac{v^2}{r}$ would give:

$$W^{\text{tot}} = \int_0^R \vec{F} \cdot d\vec{r} = \int_0^R m_0 \frac{v^2}{r} dr.$$

The definition of rigid body rotation gives the relation $\frac{v}{r} = \omega_0$, from which $\frac{dr}{r} = \frac{dv}{v}$ is deduced. Then, the classical formulation of kinetic energy theorem emerges:

$$W^{\text{tot}} = \int_0^R m_0 v dv = \frac{1}{2} m_0 v_R^2 = E_c.$$

Let us copy this calculation in the case of relativistic rigid body rotation for the force work along the space circle. According to equation (1) and figure 1, the distance r along the space circle obeys the relation $r = \Omega \frac{R}{2} = (\frac{\pi}{2} - \theta)R$, from which $dr = -R d\theta$ can be deduced. Let's note that Coriolis acceleration always perpendicular to the motion along the space circle from O to R never works and can be forgotten. The only component of acceleration in equation (1) able to give a working force is the radial projection of inertial acceleration $a_r = \frac{v^2}{r_0} \sin\theta$. This latter component perfectly compensates the relativistic mass $m = \frac{m_0}{\sin\theta}$ in such a way that the previous classical inertial force is recovered: $\frac{m_0 v^2}{\sin\theta r_0} \sin\theta = m_0 \frac{v^2}{r_0}$.

The total work of this inertial force from the center to the horizon of the relativistic rigid body rotation can be expressed as :

$$W^{\text{tot}} = \int_0^R \vec{F} \cdot d\vec{l} = \int_0^R m_0 \frac{v^2}{r_0} dr = - \int_{\frac{\pi}{2}}^0 m_0 \frac{v^2}{r_0} R d\theta.$$

From the equality $\frac{v}{c} = \frac{r_0}{R} = \cos\theta$, it straightly follows that:

$$W^{\text{tot}} = - \int_{\frac{\pi}{2}}^0 m_0 c^2 \cos\theta d\theta,$$

$$W^{\text{tot}} = m_0 c^2 \quad (4).$$

In equation (4), the famous Einstein equation that relates mass to energy can be recognized. Here it has a special interpretation; By postulating a homogeneous, isotropic 'ether', the kinetic energy theorem confers energy and mass to any relativistic rigid body rotation according to the equation:

$$E_0 = J\omega_0^2 = m_0 c^2 > 0, \text{ any rotation has a mass} \quad (5).$$

This mass equation is only the first part (a) of the ether hypothesis. It possesses a second non-independent part (b). This latter part provides a clear link between mass and rotation. Indeed, it explains the nature of the mass density, or the ether introduced in an *ad-hoc* way by the first part (a): it is the elastic energy of space-time curvature ξ_0 , a direct consequence of the velocity field. This idea naturally follows general relativity that makes a mass gravity field the origin of a space-time curvature. Reciprocal logic implies that space-time curvature should be the ultimate nature of mass. Starting from the remarkable Gauss theorem as a measurement of topological space curvature of the relativistic rigid body rotation, the total elastic energy E is introduced through a local energy ξ_0 attributed to the infinitesimal surface element dS :

$$E = \frac{1}{4\pi} \int^{\text{sphere}} \frac{\xi_0}{R^2} dS \quad (6).$$

In the relativistic rigid body rotation represented in figure 1, a standard infinitesimal surface of a sphere with a radius R would be: $dS = 2\pi R^2 \cos\alpha d\alpha$. However, on the horizon where the integral is conducted, length and surface are compressed to their extremum. Therefore, the relativistic infinitesimal surface becomes $dS = 2\pi R^2 \cos\alpha d\alpha \sin\theta_{\min}$. Using the minimum

distance of Table 1, $R \sin \theta_{\min} = r_{\text{pl}}$, the final formulation of a relativistic infinitesimal surface writes $dS = 2\pi R r_{\text{pl}} \cos \alpha d\alpha$. The total elastic energy of the relativistic rotation becomes:

$$E = \frac{\xi_0 r_{\text{pl}}}{R} \int_{\frac{\pi}{2}}^0 \cos \alpha d\alpha = \frac{\xi_0 r_{\text{pl}}}{R} \quad (7).$$

Total elastic curvature energy, or equivalently ether energy, is inversely proportional to the radius R of the relativistic rigid body rotation horizon. For a Planck particle, the maximum rotation possesses the minimum radius defined, according to the starting postulate as $r_{\min} = r_{\text{pl}}$. The elastic curvature energy is thus maximum: $E_{\max} = \xi_0$. According to equation (5), it is known that energy can also be expressed as the product of mass by the square of light speed according to: $E_{\max} = m_{\text{pl}} c^2$. Therefore, the curvature energy of equation (7) becomes: $E = \frac{m_{\text{pl}} c^2 r_{\text{pl}}}{R}$.

Another expression of the elastic curvature energy can also be retrieved. This is proportional to the rigid body rotation ω according to $E = m_{\text{pl}} c r_{\text{pl}} \omega$. Here, the quantum momentum $m_{\text{pl}} c r_{\text{pl}}$ has been introduced by Max Planck, and it is usually denoted as $\hbar$. The elastic curvature energy can be written in the following form called the photon equation :

$$E = \frac{m_{\text{pl}} c^2 r_{\text{pl}}}{R} = \hbar \omega, \quad (8).$$

This second part (b) of the ether hypothesis stipulates that any relativistic rotation has elastic curvature energy proportional to its rotation rate ω . Once again, the mass of ether, as defined in the mass equation (5), is indeed the expression of the elastic curvature of space induced by the rotation velocity field as defined in the photon equation (8). This equivalence of part (a) and (b) of the ether hypothesis, between the mass and the photon equations, that conveys a geometric origin to the mass, is the result of Stokes theorem (see appendix): any rest mass contains a curvature energy that expresses at the same time the work of inertial force along a space circle and the Coriolis induction through the same space circle. The first law of physics can thus be formulated as follows:

First law of physics

$$E = m_0 c^2 = \hbar \omega, \text{ any rotation is a mass and vice-versa} \quad (9).$$

Now, by introducing the universal gravitation constant G defined by Isaac Newton through the classical equation of gravitational radius¹ $c^2 r_{\text{pl}} = G m_{\text{pl}}$, elastic curvature energy becomes:

$$E = \frac{m_{\text{pl}} c^2 r_{\text{pl}}}{R} = \frac{G m_{\text{pl}}^2}{R} \quad (10).$$

In its present form, elastic curvature energy is the gravitational potential of a binary system of Planck particles. From the product of (8) by (10) appears a link between the constants: $G \hbar = r_{\text{pl}}^2 c^3$. A rapid check from the literature of the Planck mass and radius, $m_{\text{pl}} \sim 2.1 \cdot 10^{-8} \text{kg}$, $r_{\text{pl}} \sim 1.6 \cdot 10^{-35} \text{m}$, confirms the orders of magnitude of the gravitation constant, $G \sim 7 \cdot 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, and of $\hbar \sim 3 \cdot 10^{-34} \text{J.s}$ the Planck constant. These results are neither new nor surprising since Max Planck derived in 1899 the particle properties, r_{pl} and m_{pl} , precisely from the photon law and the gravitation law. Here, the novelty is the physical interpretation that universal theory gives to

¹ The gravitational radius is obtained by computing the satellite speed of a body in a circular orbit around a mass from equilibrium of inertial and gravitational forces: $\frac{v^2}{r} = \frac{Gm}{r^2}$. If a photon is kept in such an orbit, the gravitational radius gives: $r_G = \frac{Gm}{c^2}$. This gravitational radius should not be confused with the Schwarzschild radius that defines a black hole and is twice as big: $r_S = \frac{2Gm}{c^2}$.

the latter mathematical manipulation of fundamental constants. Indeed, from the first law of physics, relativistic rigid body rotation grants the status of fundamental particle since it becomes the elementary brick of mass. By introducing a new constant defined as $\bar{k} = m_{pl}r_{pl} = m_0R = \frac{\hbar}{c}$, the first law of physics expresses the invariance of the mass-momentum of elementary particles. Consequently, the smallest and heaviest of all particles, namely the Planck particle, has the same mass-momentum as any other elementary particle. The mass law for elementary particle, $m_0 = \frac{m_{pl}r_{pl}}{R}$, shows that any mass is a manifestation of the Planck mass, the smallest one being the universe particle $m_U = \frac{m_{pl}r_{pl}}{R_U} \approx 2.10^{-69}\text{kg}$. Finally, equation (9), $E = \hbar\omega$, can also be written as form $E = \bar{k}a$ where $a = \frac{c^2}{R}$ and $\bar{k} = \frac{\hbar}{c}$ is called the curvature constant.

Toward a fully geometric theory

A rapid check on fundamental constants shows that, in addition to the three universal cinematic constants ($v_{\max}$, $r_{\max}$, $\omega_{\max}$) of the first postulate, three physical constants, $\hbar$, G and m_{pl} , has been introduced by the first law of physics. It has been shown that G is not independent of $\hbar$. Two constants remain. They are linked by equation (9) $m_{pl} = \frac{\hbar}{cr_{pl}}$. This latter equation that relates mass to geometry thus reduces the number of additional fundamental constants brought by the first law of physics to unity. The Planck constant $\hbar$ or, equivalently but more directly, its counterpart, the curvature constant $\bar{k}$, would be a kind of intrinsic property of space that resists curvature, a property necessary for the definition of any mass. With some boldness, let us consider the possibility to define this latter intrinsic property as a direct consequence of the three fundamental geometry constants, for example as the Planck particle radius, $\hbar = r_{pl}$. Therefore, the three fundamental cinematic constants of the first postulate would remain, like the number of space dimensions. But the consequences are puzzling. This purely geometric view of physics would overturn our comprehension of physical quantities and their dimensions. The Planck mass would then become the inverse of the speed of light, $m_{pl} = \frac{2\pi}{c}$. The topologic nature of mass would be the inverse of speed, a linear mass momentum would be without any dimension, an energy would be transformed into a simple speed and the force, dear to Isaac Newton, would become a rotation, dear to René Descartes. In this purely geometrical theory, the curvature constante $\bar{k} = m_{pl}r_{pl}$ that defines the invariance of mass momentum of any elementary particle would be expressed as a Planck duration: $\bar{k} \approx 3.10^{-43}\text{s}$. In the subsequent development of this theory, the final step will not be taken, such that the four fundamental constants c , r_{pl} , R_u and $\hbar$ will be retained.

Conclusion

To conclude, the departure point of the manuscript is the relativistic rigid body rotation. It thus constitutes the postulate of a cinematic theory. Three fundamental constants ($v_{\max}$, $r_{\max}$, $\omega_{\max}$) are then introduced for symmetry reasons. This latter cinematic theory becomes a physical theory when mass is introduced through the first law of physics. This first law which relies on Stockes theorem, establishes that Planck mass is a purely geometric quantity inversely

proportional to the distance. The theory can be summarized in a lapidary formula: everything is rotation. Tests and consequences of the rotation theory will be presented in future publications starting with astrophysics.

Annexe

Through the Stokes theorem $\oint \vec{A} \cdot d\vec{r} = \iint \text{rot} \vec{A} \cdot d\vec{S}$, the link between the parts (a) and (b) of the ether hypothesis is retrieved; the work and curvature energies described in the equation of mass (5) and equation of photon (8) are equals. Indeed, when the force extracted from Eulerian equation (2) is expressed in the cylindrical coordinate system of the rotation, it becomes:

$\vec{A} = m_0(\frac{v^2}{r_0} \sin\theta \vec{u}_r + 2\frac{v^2}{r_0} \cos\theta \vec{u}_\varphi)$. Then, the first integral of the Stokes theorem gives as a result of total work along the space circle, the result of equation (4) again:

$$\oint \vec{A} \cdot d\vec{r} = -2 \int_{\frac{\pi}{2}}^0 m_0 \frac{v^2}{r_0} \sin\theta R d\theta = -2m_0 c^2 \int_{\frac{\pi}{2}}^0 \cos\theta \sin\theta d\theta = -2m_0 c^2 \left[\frac{\sin\theta^2}{2} \right]_{\frac{\pi}{2}}^0 = m_0 c^2.$$

To summarize, any rest mass contains an energy that expresses the work of inertial force along a space circle.

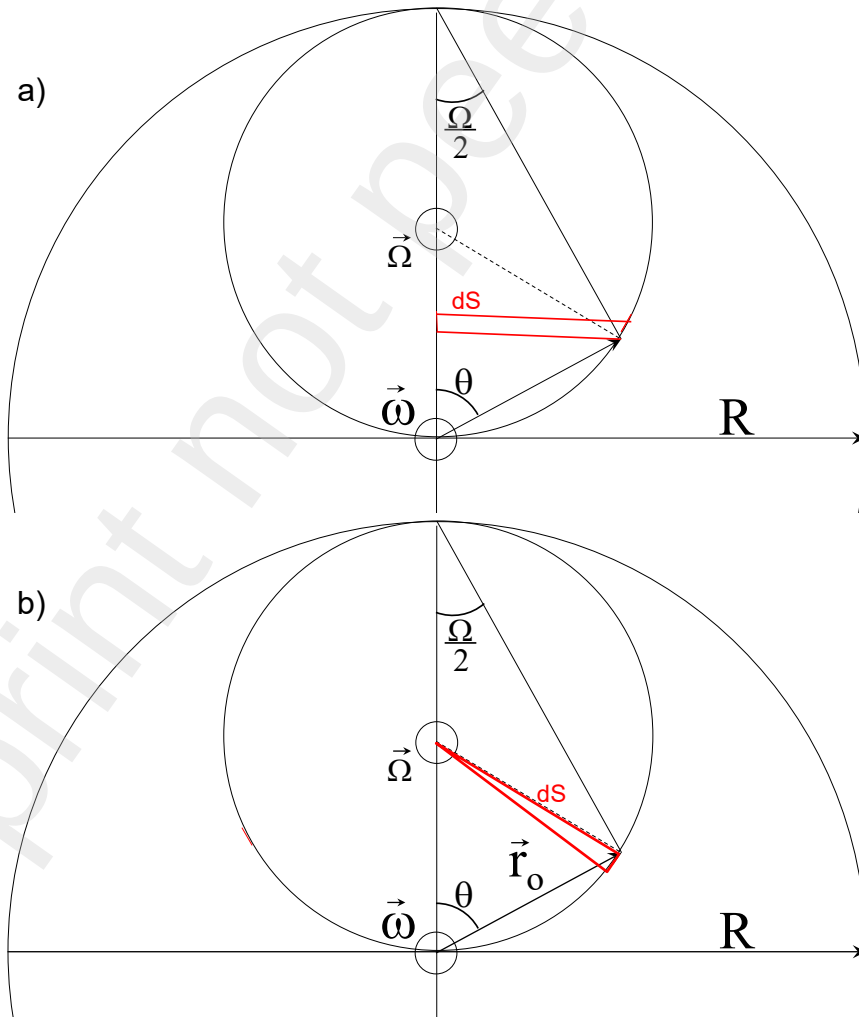


Fig 2 : a) Surface element (in red) of space circle as measured from the center of the rigid body rotation. b) Surface element (in red) of space circle as measured from its own center.

The second Stokes integral is a surface integral. It is thus naturally related to the curvature energy as defined by the Gauss theorem. However, the computations conducted in the present section are different and demands a new point of view. Let us first compute the rotational operator from the central point Ω of the space circle, figure 2b. The great interest of this point of view is to manage the Coriolis acceleration only, equation (8), the inertial acceleration $\vec{\omega} \wedge \vec{v}_\varphi$ being devoid of any rotational component. As a result, this rotational is:

$$\begin{aligned} \vec{\text{rot}}_\Omega \vec{A} &= \\ &= -\frac{2}{R} \frac{\partial}{\partial \Omega} (-2m_0 \omega v_r) \vec{u}_z = 4m_0 \omega^2 \frac{\partial}{\partial \Omega} \cos \theta^2 \vec{u}_z \\ &= -8m_0 \omega^2 \cos \theta \sin \theta \frac{\partial \theta}{\partial \Omega} \vec{u}_z = -4m_0 \omega^2 \sin 2\theta \frac{\partial \theta}{\partial \Omega} \vec{u}_z. \end{aligned}$$

By construction of figure 1, $\frac{\Omega}{2} = \frac{\pi}{2} - \theta$, it is deduced that $\frac{\partial \theta}{\partial \Omega} = -\frac{1}{2}$ on one side and further that $2\theta = \pi - \Omega$ on the other side. It follows that the rotational operator is expressed as:

$$\vec{\text{rot}}_\Omega \vec{A} = 2m_0 \omega^2 \sin \Omega \vec{u}_z.$$

By using the surface element $dS = \frac{R^2}{4} d\Omega$ described in figure Figure 2b, the second integral of the Stokes theorem becomes:

$$\iint \vec{\text{rot}}_\Omega \vec{A} \cdot d\vec{S} = \int_0^\pi 2m_0 \omega^2 \sin \Omega \frac{R^2}{4} d\Omega = -\frac{m_0 c^2}{2} [\cos \Omega]_0^\pi = m_0 c^2.$$

This result validates the first Stokes integral: any rest mass contains an energy that expresses the Coriolis induction through the space circle. Coriolis induction and inertial work are the two faces of a same energy.

One last remark remains necessary before conclusion. By changing the angle variable using $2\alpha = \Omega$, it follows that $2d\alpha = d\Omega$ and $\frac{\sin \Omega d\Omega}{4} = \sin \alpha \cos \alpha d\alpha$. Therefore, the latter integral can be expressed as:

$$2 \int_0^\pi 2m_0 \omega^2 \sin \Omega \frac{R^2}{8} d\Omega = \frac{1}{2\pi} \int_0^\pi \frac{2m_0 c^2}{R^2} 2\pi R^2 \sin \alpha \cos \alpha d\alpha = \frac{1}{2\pi} \int_0^\pi \frac{\xi_0}{R^2} dS.$$

This expression is the Gauss theorem. The second Stokes integral is thus a curvature energy. In contrast with the Gauss integral computed on the horizon and involving a multiplication by the term $\sin \theta_{\min}$, equation 7, the infinitesimal integration surface $2\pi R^2 \sin \alpha \cos \alpha d\alpha$ used in the second Stokes integral implies its Lorentz contraction along the space circle. At last, this integration circle, twice smaller than the horizon circle explains the factor of 2 between the surface energy density ξ_0 . These two approaches lead to the same final result and to the conclusion that any rest mass contains a curvature energy.

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⁵ R.C. Jennison, "Ray paths in a rotating system" Nature 199, 739, 1963.

⁶ Ø. Grøn, "Relativistic description of a rotating disk with angular acceleration", Found. of Phys. 9, 353, 1979.